NOT MEASUREMENT SENSITIVE

MIL-PRF-32587
18 January 2018

PERFORMANCE SPECIFICATION

REMOVER, PAINT, HAP-FREE, NEUTRAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for two types of paint removers that contain no hazardous air pollutants (HAPs).

1.2 Classification. The paint removers covered by this specification shall be of the following types.

Type I - For removal of epoxy primer/polyurethane topcoat and flexible polyurethane primer/polyurethane topcoat coating (paint) systems
Type II - For removal of Chemical Agent Resistant Coatings (CARC)

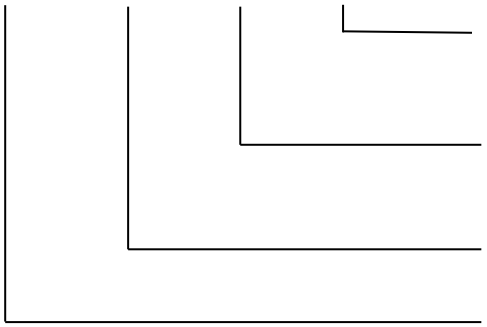
Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division Lakehurst (Code 4.1.2.2), Route 547, Joint Base MDL, NJ 08733-5100 or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3 Part identification number (PIN). Part numbers may be coded as follows:

M XXXXX - X XXXX



Container size 1/: 005G = 5 gallons
055G = 55 gallons

Type: 1 = Type I
2 = Type II

Specification number

M prefix

1/ The container size and designator may be modified for ease of procurement and is not otherwise limited (1 pint would be coded as 001P, 1 gallon would be coded as 001G, etc.).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

CCC-C-440	-	Cloth, Cheesecloth, Cotton, Bleached And Unbleached
TT-P-2760	-	Primer Coating: Polyurethane, Elastomeric, High-Solids

FEDERAL STANDARD

FED-STD-141	-	Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing

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COMMERCIAL ITEM DESCRIPTION

A-A-58054	-	Abrasive Mats, Non-Woven, Non-Metallic

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-PRF-23377	-	Primer Coatings: Epoxy, High-Solids
MIL-PRF-32348	-	Powder Coating, Camouflage Chemical Agent Resistant Systems
MIL-DTL-53022	-	Primer, Epoxy Coating, Corrosion Inhibiting Lead and Chromate Free
MIL-DTL-53039	-	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-DTL-64159	-	Camouflage Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant
MIL-DTL-81706	-	Chemical Conversion Materials for Coating Aluminum and Aluminum Alloys
MIL-PRF-85285	-	Coating: Polyurethane, Aircraft and Support Equipment
MIL-PRF-85582	-	Primer Coatings: Epoxy, Waterborne

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

PROTECTION OF THE ENVIRONMENT

40 CFR 61	-	National Emission Standards For Hazardous Air Pollutants
40 CFR 82	-	Protection of Stratospheric Ozone
40 CFR 401	-	Effluent Guidelines and Standards - General Provisions
40 CFR 796.3100	-	Aerobic Aquatic Biodegradation

(Copies of these documents are available from http://www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from www.asq.org.)

ASTM INTERNATIONAL

ASTM D92	-	Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester
ASTM D95	-	Standard Test Method for Water in Petroleum Products and Bituminous Materials by Distillation
ASTM D823	-	Standard Practices for Producing Films of Uniform Thickness of Paint, Varnish, and Related Products on Test Panels.
ASTM D2196	-	Standard Test Methods for Rheological Properties of Non-Newtonian Materials by Rotational Viscometer
ASTM E70	-	Standard Test Method for pH of Aqueous Solutions With the Glass Electrode
ASTM F483	-	Standard Practice for Total Immersion Corrosion Test for Aircraft Maintenance Chemicals
ASTM F519	-	Standard Test Method for Mechanical Hydrogen Embrittlement Evaluation of Plating/Coating Processes and Service Environments
ASTM F1080	-	Standard Test Method for Determining the Consistency of Viscous Liquids Using a Consistometer
ASTM F1110	-	Standard Test Method for Sandwich Corrosion Test

(Copies of these documents are available from www.astm.org.)

NATIONAL TOXICOLOGY PROGRAM

Annual Report on Carcinogens

(Copies of this voluntary standard are available from the Annual Report on Carcinogens, National Toxicology Program, PO Box 12233, Research Triangle Park, NC 27709.)

SAE INTERNATIONAL

SAE AMS2759/3	-	Heat Treatment Precipitation-Hardening Corrosion-Resistant and Maraging Steel Parts
SAE AMS4049	-	Aluminum Alloy, Sheet and Plate, Alclad 5.6Zn - 2.5Mg - 1.6Cu - 0.23Cr (Alclad 7075; -T651 Plate) Solution and Precipitation Heat Treated
SAE AMS4911	-	Titanium Alloy, Sheet, Strip, and Plate, 6Al - 4V, Annealed

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SAE AMS5046	-	Carbon Steel, Sheet, Strip, and Plate (SAE 1020 and 1025) Annealed. (DoD Adopted)
SAE AMS5548	-	Steel, Corrosion and Heat Resistant, Sheet and Strip 16.5Cr 4.5Ni 2.9Mo 0.10N, 1850 to 1975°F (1010 to 1079°C), Annealed
SAE AMS5629	-	Steel, Corrosion-Resistant, Bars, Wire, Forgings, Rings, and Extrusions, 13Cr - 8.0Ni - 2.2Mo - 1.1Al, Vacuum Induction Plus Consumable Electrode Melted, Solution Heat Treated, Precipitation Hardenable
SAE AMS6359	-	Steel, Sheet, Strip, and Plate 0.80Cr - 1.8Ni - 0.25Mo (0.38 - 0.43C) (SAE 4340)
SAE AMS6512	-	Steel, Maraging, Bars, Forgings, Tubing, and Rings 18Ni - 7.8Co - 4.9Mo - 0.40Ti - 0.10Al Consumable Electrode Vacuum Melted, Annealed
SAE AMS-QQ-A-250/4	-	Aluminum Alloy 2024, Plate and Sheet. (DoD Adopted)
SAE AMS-QQ-A-250/5	-	Aluminum Alloy Alclad 2024, Plate and Sheet. (DoD Adopted)
SAE AMS-QQ-A-250/12	-	Aluminum Alloy 7075, Plate and Sheet
SAE AMS-QQ-P-416	-	Plating, Cadmium (Electrodeposited)
SAE AMS-H-6875	-	Heat Treatment of Steel Raw Materials

(Copies of these documents are available from www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The paint removers furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Toxicity. The paint remover shall not affect the health of personnel when used for its intended purpose. Prior to listing on the qualified products list, the manufacturer shall obtain a satisfactory toxicity review (see 4.2.2.1).

3.3 Compositional limitations. The remover shall be biodegradable and shall not contain Hazardous Air Pollutants (HAPs) (as defined in 40 CFR 61), known or suspected human carcinogens (as defined by the National Toxicology Program's Annual Report on Carcinogens), toxic pollutants (as defined in 40 CFR 401), nor Class I or II Ozone Depleting Substances (as defined in 40 CFR 82). Surface active agents used in the remover shall be not less than 90

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percent biodegradable in 28 days, as determined in accordance with 40 CFR 796.3100 (see table I).

3.3.1 Paint remover pH level. The paint remover shall have a pH of no less than 5.0 and no more than 8.0, when tested in accordance with ASTM E70 (see table I).

3.3.2 Water content and alkalinity. The water content and alkalinity of the remover shall be determined during the qualification inspection in accordance with ASTM D95 and 4.5.1 of this specification, respectively. Conformance inspection results for water content and alkalinity (see 4.3) shall not differ from the manufacturer's certified target values (see 4.2.2) by more than ±10.0 percent of the value.

3.4 Flash point. The flash point of the remover shall be not less than 60 °C (140 °F), when tested in accordance with ASTM D92.

3.5 Consistency. The remover shall flow 100 to 230 millimeters (mm) from the origin, when tested in accordance with ASTM F1080.

3.6 Viscosity. The viscosity of the remover shall be 6 to 12 Pascal-seconds (Pa-sec) [6,000 to 12,000 centipoise (cP)] at 30 revolutions per minute (rpm), when tested in accordance with ASTM D2196, method A, and 4.5.2. Additionally, the remover shall remain wet on a vertical surface for not less than 6 hours.

3.7 Paint removal. The stripping percentage shall be equal to, or better than, the stripping percentage obtained with the control formula, when tested in accordance with 4.5.3.

3.8 Corrosivity.

3.8.1 Immersion corrosion. The remover shall not cause any visible corrosion (staining is allowable on cadmium plated specimens) nor an average weight change of any specimen greater than that shown in table V (for type I paint removers) and table VI (for type II paint removers), when tested in accordance with 4.5.4.

3.8.2 Sandwich corrosion. The remover shall have a sandwich corrosion rating no greater than one, or no effect greater than that obtained for reagent water, when tested in accordance with ASTM F1110. Additionally, when viewed under 10X magnification, the test specimen shall exhibit no white aluminum corrosion product.

3.8.3 Hydrogen embrittlement. Type I remover shall not cause embrittlement of high strength steel when tested on ASTM F519 Type 1a (notched round bar) specimens in accordance with 4.5.5 (1). Type II remover shall not cause embrittlement of high strength steel when tested on ASTM F519 Type 1d (notched C-ring) specimens in accordance with 4.5.5 (2).

3.9 Accelerated storage stability. The remover shall be conditioned in accordance with 4.5.6. Upon completion of the conditioning, the remover shall not exhibit any precipitation,

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layering, separation (not greater than 5 percent by volume separation), nor marked change in color (as compared to its original color); additionally, the remover shall be mixable and shall conform to 3.5, 3.6, and 3.7.

3.10 Rinseability. There shall be no visible remover residue, when tested in accordance with 4.5.7.

3.11 Storage stability. After a storage time of not less than 6 months (see 4.5.8), the remover shall meet all of the requirements of this specification, with the exception of 3.13 (service evaluation).

3.12 Workmanship. The remover shall have a uniform and homogeneous appearance and the component ingredients shall be blended and processed.

3.13 Service evaluation. The remover shall pass an aircraft production stripping facility process for removal of the intended primer/topcoat system (see table III) from aircraft exteriors, when tested in accordance with 4.5.9.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 Qualification inspection. The qualification inspection shall consist of all the tests specified in table I. Upon successful completion of all laboratory tests, the service evaluation test shall be performed as part of the qualification inspection.

4.2.1 Qualification samples. Prior to submitting samples for qualification testing, vendors shall request authorization from the qualification activity (see 6.3). Upon receipt of authorization, samples shall be forwarded as directed. The qualification test sample shall be contained in one 3.8 liter (one gallon) glass container and one 3.8 liter (one gallon) unit in the type of container in which the manufacturer intends to supply contract quantities of the remover. (If the manufacturer intends to supply contract quantities of the remover in plastic, the second qualification sample shall be made from the same type plastic and have no greater wall thickness than containers proposed for production packaging.) The Safety Data Sheet (SDS) (see 3.2 and 6.4) shall accompany each submission. Samples shall be identified as follows:

Qualification test samples
Performance Specification MIL-PRF-32587, "REMOVER, PAINT, HAP-FREE, NEUTRAL"
Manufacturer's name and product number
Batch or lot number

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Date compounded
Submitted by (name and date) for qualification testing in accordance with authorization (reference authorizing letter)

4.2.1.1 Field service test samples. Upon successful completion of all laboratory tests, the manufacturer will be authorized to submit samples for a field service evaluation. Samples shall be forwarded in accordance with instructions contained in the authorizing letter and shall consist of 380 to 2840 liters (100 to 750 gallons) of paint remover, as required by the field service activity and stated in the authorizing letter. No overpacking is required.

TABLE I. Qualification test methods.

Characteristic	Requirement Paragraph	Test Method or Paragraph
Biodegradability	3.3	40 CFR 796.3100
pH	3.3.1	ASTM E70 1/
Water content	3.3.2	ASTM D95
Alkalinity	3.3.2	4.5.1
Flash Point	3.4	ASTM D92
Consistency	3.5	ASTM F1080
Viscosity	3.6	ASTM D2196 & 4.5.2
Paint removal	3.7	4.5.3
Immersion corrosion	3.8.1	4.5.4
Sandwich corrosion	3.8.2	ASTM F1110
Hydrogen embrittlement	3.8.3	4.5.5
Accelerated storage stability	3.9	4.5.6
Rinseability	3.10	4.5.7
Storage stability	3.11	4.5.8
Workmanship	3.12	2/
Service evaluation	3.13	4.5.9

1/ Mix 50 ml of remover with 50 ml of distilled water and then test.
2/ Workmanship observed throughout qualification inspection.

4.2.2 Qualification data. In addition to the qualification test samples, the manufacturer shall furnish the following to the qualification activity:

a. One copy of the SDS (see 3.2);
b. A certified test report showing that the material conforms to the requirements of this specification. Test(s) not conducted due to lack of special facilities or materials shall be noted in the report;
c. Certification of compliance with the requirements for toxicity (see 3.2), compositional limitations (see 3.3), and target values for water content and alkalinity (see 3.3.2); and

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d. Certification that the remover is biodegradable, with details concerning the type of tests conducted and their results for each of the surfactants in the product formula.

4.2.2.1 Toxicity review. The contractor shall furnish the toxicological data and formulations required to evaluate the safety of the material for the proposed use. When requested by the qualifying activity, the manufacturer of the product shall provide the following information:

a. Product name (including part number/tradename), formula, CAS number, and percentage by weight of each ingredient in the product;
b. Product SDS compliant with OSHA Form 174 (see 6.4);
c. Current SDS for each ingredient used in the formulation; and
d. The results of any toxicological testing of the product; identification of its pyrolysis products; and any other information as may be needed to permit an accurate appraisal of any toxicity problem or issues associated with the handling, storage, application, use, removal, disposal, or combustion of the material.

Information submitted shall be marked to show it is being provided in connection with qualification under this specification.

4.2.3 Retention of qualification. In order to retain qualification of products approved for listing on the qualified products list (QPL), the manufacturer shall verify by certification to the qualifying activity that the product complies with the requirements of this specification. The time of periodic verification by certification shall be two-year intervals from the date of original qualification and shall be initiated by the Government. The Government reserves the right to reexamine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specification requirements.

4.3 Conformance inspection.

4.3.1 Batch and lot formation. A batch shall consist of all material manufactured during one continuous operation and forming part of one contract or order for delivery. A lot shall consist of all material manufactured at one time from one batch, forming part of one contract, and submitted for acceptance. When requested by the acquisition activity (see 6.2), the manufacturer shall furnish with each batch, a certified test report showing that the material has passed the conformance inspection and that there has been no formulation or process change from that which resulted in the production of the qualification inspection sample.

4.3.2 Sampling.

4.3.2.1 Sampling for conformance inspection. Conformance test samples shall be selected in accordance with FED-STD-141, method 1022, with the exception that ANSI/ASQ Z1.4 shall be used for sampling; the sample shall consist of not less than 3.8 liters (1 gallon). If required (see 6.2), the manufacturer shall certify that the material has been manufactured in the same

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manner and using the same base ingredients as the approved qualification sample. The selected samples shall be tested as specified in 4.3.3.1.

4.3.2.2 Sampling for inspection of filled containers. A random sample of filled containers shall be selected from each lot in accordance with ANSI/ASQ Z1.4, Inspection Level I, for the inspection specified in 4.3.3.2. For the purposes of this inspection, the lot size shall be the number of shipping containers.

4.3.3 Inspections.

4.3.3.1 Physical tests. The sample selected in 4.3.2.1 shall be tested for conformance to table II.

4.3.3.2 Filled container inspection. The sample selected in accordance with 4.3.2.2 shall be visually examined for fill, closure, and marking.

4.4 Inspection conditions. Unless otherwise specified in the test method or paragraph, all tests shall be conducted at standard conditions (room temperature of 21 ±3 °C [70 ±5 °F], and relative humidity of 50 ±10 percent).

TABLE II. Conformance inspections.

Characteristic	Requirement Paragraph	Test Method or Paragraph
pH	3.3.1	ASTM E70 1/
Water content	3.3.2	ASTM D95
Alkalinity	3.3.2	4.5.1
Consistency	3.5	ASTM F1080
Viscosity	3.6	ASTM D2196 & 4.5.2
Paint Removal	3.7	4.5.3
Immersion corrosion	3.8.1	4.5.4
Hydrogen embrittlement	3.8.3	4.5.5

1/ Mix 50 ml of remover with 50 ml of distilled water and then test.

4.5 Test methods. The tests of this specification shall be conducted in accordance with table I and 4.5.1 through 4.5.9.

4.5.1 Alkalinity. Prepare and standardize 1 Normal (N) hydrochloric acid (normality = A) and 1 N sodium hydroxide (normality = B). Solutions at 0.1 N may be used when products do not have a high reserve alkalinity. With a syringe, transfer 10 grams of remover into a 250 ml beaker containing 75 ml of methanol and 50.0 ml of the 1 N hydrochloric acid. Determine the sample weight (S) by difference. Add a stirring bar, then mix and titrate with the 1 N sodium

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hydroxide to pH 4.0. Determine the volume (in milliliters) of titrant used (V) and calculate alkalinity as follows:

Alkalinity (in percent NH_3 (28 percent)) = [(50A - VB) x 6.07 ÷ S]

4.5.2 Viscosity. A Brookfield viscometer (Model LVF or similar) shall be used with spindle number LV4. Viscosity shall be determined in accordance with ASTM D2196, method A, at 30 rpm, but only after the spindle has been operating for at least 3 minutes. In addition, prepare a test panel with the coating system specified in table III, in accordance with 4.5.3.1 and 4.5.3.2. Apply remover to the test panels and place them on a rack, such that the surface forms a 90° angle with the horizontal and store for six hours. At the end of this storage period, wipe a gloved finger across the center of the vertical test panel surface; visual inspection shall reveal an obvious streak in the remover.

4.5.3 Paint removal. Not less than four test panels for each coating system (solvent-borne primer system, water-borne primer system, and flexible polyurethane primer system) shall be tested; two for the control and two for the remover under test.

4.5.3.1 Preparation of test panels. Test panels, measuring approximately 0.5 x 127.0 x 406.4 mm (0.020 x 5.0 x 16.0 in.) and constructed of deburred Alclad aluminum alloy 2024 conforming to SAE AMS-QQ-A-250/5, shall be prepared for application of coating as follows:

a. Abrade the surface of the test panel with a very fine abrasive mat, conforming to A-A-58054, which has been soaked with deionized water, by manually rubbing the mat back and forth parallel to the long dimension of the panel until the entire surface is water break free;
b. Immediately wipe the test panel clean and dry with cotton cloth conforming to CCC-C-440, Class 1;
c. Within 4 hours of drying, immerse the panel in chemical conversion coating, conforming to MIL-DTL-81706, Type I, Class 1A, for three minutes. Ensure that panels do not touch each other or the sides of the tank;
d. Remove the test panels and rinse thoroughly with deionized water for one minute; and
e. Allow test panels to air dry in an upright position and, within 24 hours, apply the coating system from table III for which the remover is intended.

4.5.3.2 Application of coatings to test panels. Apply the applicable coating system from table III to the test panels. MIL-PRF-23377, MIL-PRF-85582, TT-P-2760, and MIL-PRF-85285 shall be applied in accordance with ASTM D823.

4.5.3.3 Scribing. Cut test panels, prepared in accordance with 4.5.3.1 and 4.5.3.2, into pieces measuring approximately 102 x 127 mm (4 x 5 in.). Scribe an "X" one inch in length in the middle of the panel on the coated surface side of each panel piece using a stylus. Ensure that the scribe cuts through to the substrate.

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4.5.3.4 Preparation of control formulation remover. Prepare the control formula remover by mixing the ingredients listed in table IV as follows in a high-speed blender. NOTE: This formula is not intended to meet any other requirements of this specification.

Mixing instructions for 500 or 1000 gram batch:

a. Weigh out Anisole.
b. Weigh out Methocel on filter paper and slowly add to the Anisole while stirring with an impeller type mixer. There shall be no lumps.
c. Weigh benzyl alcohol into the above mix and stir at full speed until homogeneous.
d. Weigh out water into a separate container.
e. Weigh 28 percent ammonia into the water and stir for a few seconds with a spatula.
f. Add 50 gram increments of the mixture from (e) to the mixture from (c) while blending at high speed. Blend after each addition to achieve a smooth milkshake consistency. Complete the formula preparation in less than 5 minutes to avoid excessive loss of ammonia.

4.5.3.5 Procedure. Seal edges of prepared test panels with beeswax by dipping in melted wax to a depth not to exceed 6 mm (0.25 in.) from all edges. Place four test panels on a rack, coated and scribed surface up, such that the test panel forms a 60° angle with the horizontal. Apply the remover by pouring along the top edge of two test panels. Immediately apply the control formula remover (see 4.5.3.4) in a similar manner to the remaining two test panels. Allow the remover to flow down the panels, taking no more than one minute to cover the coated surface. Allow the remover to dwell on the panel for 4 hours. Test panels shall then be scraped with a rubber scraper to remove loosened coatings; apply additional remover to cover any remaining coating and allow it to dwell for an additional 4 hours. Immediately after this exposure, the test panels shall be scraped with a rubber scraper to remove the bulk of the loosened coatings and remover residue. The test panels shall then be rinsed with tap water and brushed with a soft, nylon bristle brush. Paint removal shall be determined for each test panel by estimating the percentage of aluminum substrate revealed by the stripping process. The result shall be reported as the average of the two panels under test and the average of the two panels for the control formula.

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TABLE III. Test panel finishes.

Remover type (see 1.2)	Coating system	Step	Material	Number of coats and total dry film thickness	Drying time between coats	Drying time before baking	Baking time and temperature
I	Solventborne epoxy primer/polyurethane topcoat system	1	MIL-PRF-23377, Type I, Class C	One coat 0.6 to 0.9 mils (15 to 23 :m)	60 minutes	---	---
		2	MIL-PRF-85285, Type I, Class H (Color 36375)	Two coats 1.7 to 2.3 mils (43 to 58 :m)	15 minutes	7 days	7 days at 150 °F
	Waterborne epoxy primer/polyurethane topcoat system	1	MIL-PRF-85582, Type I, Class C2	One coat 0.6 to 0.9 mils (15 to 23 :m)	60 minutes	---	---
		2	MIL-PRF-85285, Type I, Class H (Color 36375)	Two coats 1.7 to 2.3 mils (43 to 58 :m)	15 minutes	7 days	7 days at 150 °F
	Flexible polyurethane primer/polyurethane topcoat system	1	TT-P-2760, Type I, Class C	Two coats 1.5 - 2.0 mils (38 to 51 :m)	60 minutes	---	---
		2	MIL-PRF-85285, Type I, Class H (Color 36375)	Two coats 1.7 to 2.3 mils (43 to 58 :m)	15 minutes	7 days	7 days at 150 °F
II	High Solids Epoxy Primer/Powder CARC	1	MIL-PRF-23377, Type I, Class N	One Coat 0.6 to 0.9 mils (15 to 23µm)	120 min at 70°F	N/A	N/A
		2	MIL-PRF-32348, Type II, Class 2	One Coat 1.8 to 2.2 mils (46 to 56µm)	N/A	N/A	Cure per manufacturers requirements
	Corrosion Inhibiting Epoxy Primer/Powder CARC	1	MIL-DTL-53022, Type IV	One Coat 1.3 to 1.7 mils (33 to 43µm)	60 min at 70°F	N/A	N/A
		2	MIL-PRF-32348, Type III, Class 2	One Coat 3.0 to 3.5 mils (76 to 89µm)	N/A	N/A	Cure per manufacturers requirements

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TABLE III. Test panel finishes - Continued.

II	High Solids Epoxy Primer/Aliphatic Polyurethane CARC	1	MIL-PRF-23377, Type I, Class N	One Coat 0.6 to 0.9 mils (15 to 23µm)	120 min at 70°F	N/A	N/A
		2	MIL-DTL-53039	One Coat 1.8 to 2.2 mils (46 to 56µm)	N/A	Dry to handle in 30 minutes	2 hours at 180°F then at 70°F for 2 days
	Corrosion Inhibiting Epoxy Primer/Aliphatic Polyurethane CARC	1	MIL-DTL-53022, Type IV	One Coat 1.3 to 1.7 mils (33 to 43µm)	60 min at 70°F	N/A	N/A
		2	MIL-DTL-53039	One Coat 1.8 to 2.2 mils (46 to 56µm)	N/A	Dry to handle in 30 minutes	2 hours at 180°F then at 70°F for 2 days
	High Solids Epoxy Primer/Water Dispersible Aliphatic Polyurethane CARC	1	MIL-PRF-23377, Type I, Class N	One Coat 0.6 to 0.9 mils (15 to 23µm)	120 min at 70°F	N/A	N/A
		2	MIL-DTL-64159	1.8 to 2.2 mils (46 to 56µm)	N/A	Dry at 70°F for 8 hours	Bake for 2 hours at 180°F
	Corrosion Inhibiting Epoxy Primer/Water Dispersible Aliphatic Polyurethane CARC	1	MIL-DTL-53022, Type IV	One Coat 1.3 to 1.7 mils (33 to 43µm)	60 min at 70°F	N/A	N/A
		2	MIL-DTL-64159	One Coat 1.8 to 2.2 mils (46 to 56µm)	N/A	Dry at 70°F for 8 hours	Bake for 2 hours at 180°F

TABLE IV. Control formula remover.

Ingredient	Weight percent
Anisole	19.0
Methocel, grade F4M 1/	1.32
Benzyl alcohol	41.7
Water	33.2
Ammonia (28 percent)	4.78

1/ Dow Chemical

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4.5.4 Immersion corrosion. The immersion corrosion test shall be performed as specified in ASTM F483, except that preparatory solvent cleaning of test panels shall be done with acetone followed by isopropyl alcohol. The duration of immersion shall be 7 days. Weight change results shall be reported in units of mg/cm^2/24 hours. Failure to meet the limitations in table V (for type I paint removers) and table VI (for type II paint removers) is cause for rejection.

TABLE V. Immersion corrosion limits (type I removers).

Test panel	Specification	Surface Treatment	Maximum Weight Change (mg/cm^2/24 hour)
Alclad aluminum alloy 2024 (T3 temper)	SAE AMS-QQ-A-250/5	None	0.04
Alclad aluminum alloy 7075 (T6 temper)	SAE AMS4049	None	0.04
Alclad aluminum alloy 2024 (T3 temper)	SAE AMS-QQ-A-250/4	Anodize in accordance with MIL-A-8625, Type I or II	0.04
Steel 1020	SAE AMS5046	Polished to 65 root mean square (rms).	0.04
Steel 1020	SAE AMS5046	Cadmium plated in accordance with SAE AMS-QQ-P-416, Type I, Class 3	0.20
Titanium 6Al-4V	SAE AMS4911	None	0.04

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TABLE VI. Immersion corrosion limits (type II removers).[1/2/]

Test Panel	Specification and Temper	Surface Treatment	Maximum Weight Change (mg/cm^2/24 hour)
AM-355 CRT, UTS approx. 220 (180 yield) KSI, Sheet/Strip	Strap Pack Material, Cold Rolled and Tempered (ref SAE AMS-5548)	None	0.28
PH 13-8 Mo (H1000 condition, UTS approx. 205 KSI, Bar/Rod[2/]	SAE AMS5629 and H1000 condition	None	2.71
Maraging C-250 (aged 480°C/896°F for 3 hrs, approx. UTS 260 KSI, Bar/Rod[2/]	UNS K92890* and heat treat per SAE AMS-2759/3 Aged 480 °C (900 °F) for 3 hours *See SAE AMS6512 for the specific chemistry for UNS K92890	None	2.82
Al 7075-T6 (Bare, UTS approx. 83 KSI, Sheet/Strip	SAE AMS-QQ-A-250/12 and Temper T6	None	0.98
Al 2024-T3 (Bare, UTS approx. 65 KSI), Sheet/Strip	SAE AMS-QQ-A-250/4 and Temper T3	None	0.97
Ti 6Al-4V (Solution treated and aged. UTS approx. 160 KSI), Sheet/Strip	SAE AMS4911	Solution treated and aged	0.62
Steel 4340 (Oil quench from 800°C/1472°F, tempered 595°C/1100°F, UTS approx. 160 KSI, Sheet/Strip	SAE AMS6359 / SAE AMS-H-6875	Cadmium plated per SAE AMS-QQ-P-416, Type II, Class I	2.73

1/ Average of three panels
2/ Cut oversize coupons from bar or rod and grind to finish dimensions

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4.5.5 Hydrogen embrittlement. This test shall be conducted in accordance with ASTM F519, except as noted below, using specimens conforming to Types 1a (for Type I Remover) or 1d (for Type II Remover). Specimens shall be cadmium plated in accordance with ASTM F519, Table 2, treatment B at 645 amps per square meter (60 amps per square foot). Plating shall cover the notch and surfaces within 12 mm (0.5 in.) of the notch; threaded surfaces or ends of the specimens shall not be plated. Plated specimens shall be baked for 100 hours at 190 ±13 °C (375 ±25 °F).

Not less than three specimens shall be used. If failure occurs in one of the three specimens (for either type), a second set of three specimens shall be evaluated. Not less than three out of three, or five out of six specimens, shall meet the 200 hour duration for Type 1a, or 100 hour duration for Type 1d, to meet the requirement and consider the paint remover non-embrittling.

(1) Type 1a (notch round bar) specimens

(a) Specimens shall be loaded to 75 percent of the tensile notched fracture strength (NFS).

(b) After loading the specimen, the paint remover shall be applied once to the notch of each specimen using a clean, nylon bristle brush to completely cover the notch and surfaces within 12 mm (0.5 in.) of the notch.

(c) The test shall be run for not less than 200 hours; the time to failure shall be recorded if failure occurs in less than 200 hours.

(2) Type 1d (notch C-ring) specimens

(a) Specimens shall be loaded to 65 percent of the bend NFS.

(b) After loading the specimens, the stressed C-rings shall be immersed in the remover to a depth of approximately 10 mm (0.4 in.) for 60 seconds with the notched side down. The rings shall be removed and allowed to drain; no rinse of any kind shall be used after immersion.

(c) The ring shall be hung with the notched side down for not less than 100 hours; the time of failure shall be recorded if failure occurs in less than 100 hours.

4.5.6 Accelerated storage stability. A 150 ml portion of well mixed paint remover shall be poured into each of two clean 473 ml plastic bottles which shall be approximately 180 mm (7 in.) in height and 70 mm (2.8 in.) in outside diameter. One bottle shall be sealed with a screw-type cap and stored for at least six days at room temperature for reference purposes. The other bottle shall be sealed in the same manner and thoroughly shaken for 10 seconds; place in a bath maintained at 60 ±2 °C (140 ±4 °F) for 5 hours, then remove and allow to cool to ambient for 19 hours. This heating-cooling cycle shall be repeated 5 times. The last cooling cycle may be shortened to 3 hours to allow the test to be started on Monday and completed on Friday. The test

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may be allowed to run over a weekend by leaving the sample at room temperature from Friday afternoon to Monday morning. After completion of 5 cycles, both bottles shall be thoroughly shaken for 10 seconds, then allowed to remain undisturbed at room temperature for one hour. The bottles shall be examined for separation, precipitation, layering, and color change.

4.5.7 Rinseability. Approximately 2.0 grams of paint remover shall be added to a 60.0 mm (2.5 in.) diameter disposable aluminum dish. The dish shall be placed in an air-circulating oven at 49 ±2 °C (120 ±4 °F) for 24 hours, then rinsed for one minute under flowing tap water adjusted to 35 ±2 °C (95 ±4 °F). A soft horse hair or hog bristle brush with bristles no less than 20.0 mm (0.8 in.) long can be used to assist the rinsing process. The dish shall then be lightly wiped with paper tissue, dried for 10 minutes in the same oven, then examined for evidence of visible residue.

4.5.8 Storage stability. A 3.8 liter (1.0 gallon) container filled with remover furnished for storage stability shall be stored for not less than 6 months at 21 ±3 °C (70 ±5 °F). The container shall be constructed of the same material in which the manufacturer intends to supply contract quantities of the remover and have no greater wall thickness than containers proposed for production packaging. After the 6 month storage period, the remover shall be tested for all requirements in section 3, except storage stability (see 3.11) and service evaluation test (see 3.13).

4.5.9 Service evaluation test. The service evaluation shall be performed at an aircraft paint stripping site designated by the qualifying activity. The remover shall be spray applied to the clean, dry exterior surface of an aircraft which has been in service for at least one year. The remover shall be applied and allowed to act for not less than 6 hours at not less than 21 °C (70 °F). If the remover under test fails to strip effectively, it shall be compared to a product qualified to this specification and designated by the qualification laboratory. The paint remover under test and the comparison product shall be applied to alternating areas on each side of the same aircraft. The areas completely stripped by the remover under test shall be not less than 90 percent of the area stripped by the comparison product or the product under test shall require no more than 20 percent more time to achieve the same degree of stripping.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The paint remover covered by this specification is intended for stripping paint from metal surfaces (with the exception of magnesium alloys) of aircraft exteriors. Removers are used to strip epoxy primer coating/polyurethane topcoat systems and flexible polyurethane primer/polyurethane topcoat systems (see table III) and may be useful for removing self-priming polyurethane coating (TT-P-2756) systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification, including any amendments.
b. Quantity of remover desired.
c. Test/inspection reports and certifications required (4.3.1 and 4.3.2.1).
d. Packaging requirements (see 5.1).
e. Address(es) where SDSs should be sent.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-32587, whether or not such products have actually been so listed by that date. The attention of the supplier is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command (AIR-4.3.4), and information pertaining to qualification of products may be obtained from the Commander, Naval Air Warfare Center Aircraft Division, Attn.: Code 4.3.4.2, Building 2188, 48066 Shaw Rd, Unit 5, Patuxent River, MD 20670-1908.

6.4 Safety Data Sheet (SDS). The Safety Data Sheet (SDS) for the remover will be prepared in accordance with FED-STD-313 and will conform to 29 CFR 1910.1200. The 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers will identify the activities requiring copies of the SDS.

6.5 Subject term (key word) listing.

Aircraft exterior coating removal
Methyl ethyl ketone (MEK)
Stripper

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CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – AS
Air Force – 11
DLA – GS

Preparing activity:
Navy – AS
(Project 6850-2016-001)

Review activities:
Army – MI, MR
Navy – CG
Air Force – 68, 184

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil.

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